

SMART INDIA HACKATHON 2026

TITLE PAGE



Problem Statement Title: **SHRUTI** — an ultra-low-cost handheld newborn hearing screener using Distortion-Product Otoacoustic Emissions (DPOAE), to unblock universal screening under NPPCD & RBSK

PS Category- Software/Hardware: **HARDWARE** (hardware + software)

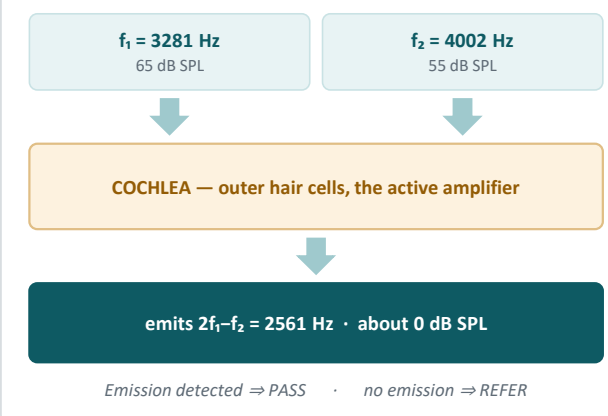
Theme: MedTech / HealthTech · Student Innovation Category

Team Name: **STATIC**

Team Members Names:

1. Abhinav Garg
2. Neha Sharma
3. Utkarsh Gupta
4. Aditya Garg
5. Ansh Goyal
6. Shruti Shreshtha

HOW IT WORKS



5–6 / 1,000

Indian newborns are hard of hearing — most are not identified until age 2 or later

₹5,000

bill of materials, against ₹1.85–2.9 lakh for an imported OAE screener — 37–58× cheaper

< 30 seconds

PASS / REFER per ear, run by an ASHA or ANM after about two hours of training

IDEA TITLE



◆ Proposed Solution (Describe your Idea/Solution/Prototype)

The physics: a healthy ear does not just receive sound — it **makes** sound. Play two pure tones into a working cochlea and its outer hair cells generate a third tone at $2f_1 - f_2$. No emission $\Rightarrow$ no working cochlea.

Detailed explanation of the proposed solution

- Handheld probe delivers two pure tones ($f_2/f_1 = 1.22$, $L_1/L_2 = 65/55$ dB SPL) at $f_2 = 2, 3, 4, 5$ kHz
- A low-noise microphone in the same probe listens for the cochlear emission at $2f_1 - f_2$ (about 0 dB SPL)
- 24-bit codec + ESP32-S3 run in-situ calibration, coherent averaging and artifact rejection on-device
- Verdict on an OLED in under 30 s: PASS / REFER / RETEST — never a diagnosis
- Result syncs over BLE to an offline-first Android app used by the ASHA or ANM

How it addresses the problem

- Cost is the documented barrier under NPPCD — SHRUTI removes it, 37–58× cheaper
- Runs without an audiologist, in a country with roughly 1 audiologist per 5,00,000 people
- Detection moves from ~24 months to the first weeks, inside the window where intervention still works
- Every REFER is auto-attached to the infant's next immunisation visit (6/10/14 weeks), attacking loss-to-follow-up
- Battery-powered and pocket-sized, so screening reaches outreach camps and home births

Innovation and uniqueness of the solution

- TWO separate balanced-armature receivers — one tone each, so the device physically cannot manufacture the emission it is looking for
- Bin-exact stimulus: f_1 , f_2 and $2f_1 - f_2$ land on exact FFT bin centres $\rightarrow$ zero spectral leakage from the 65 dB primaries
- Per-ear in-situ calibration corrects for canal volume and seal before every test
- Abstains rather than guesses: refuses to PASS on a bad seal or excess noise

TODAY

- ₹1,85,000 – ₹2,90,000 imported OAE screener
- Operated by an audiologist; tied to a hospital
- Only 38% of medical colleges run a screening programme
- Refer-and-hope: no tracking of who actually returns
- Detection typically at 24+ months — after the language window

WITH SHRUTI

- ₹5,000 bill of materials (₹2,700 at 10,000 units)
- Operated by an ASHA / ANM after ~2 hours of training
- One device per delivery point, and it can leave the building
- Every REFER tracked to an outcome, with an escalation ladder
- Detection in the first weeks — inside the window that matters

STATIC

Technologies to be used

(programming languages, frameworks, hardware)

HARDWARE

- 2 × Knowles ED-29689 balanced-armature receivers (one per tone)
- Low-noise mic, ≤ 27 dB(A) EIN (Knowles FG-23629 / ICS-40730)
- TI TLV320AIC3204 24-bit codec — DAC+ADC on one MCLK domain
- ESP32-S3-WROOM-1 N16R8 · OPA1662 preamp · 4-layer PCB
- 3D-printed 3-lumen probe, disposable silicone tips (₹5)

FIRMWARE & DSP

- C / C++ on ESP-IDF; ESP-DSP FFT, 48 kHz, 8192-point
- Phase-locked A/B coherent averaging; adaptive frame rejection

TECHNICAL APPROACH

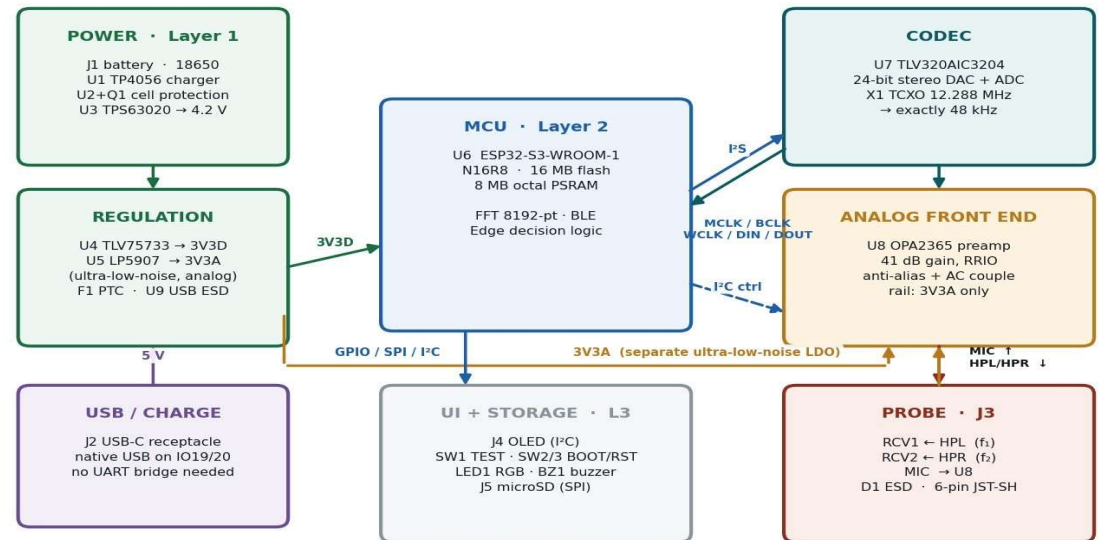
Methodology and process for implementation

(flow charts / images / working prototype)



SHRUTI · Main PCB Block Diagram

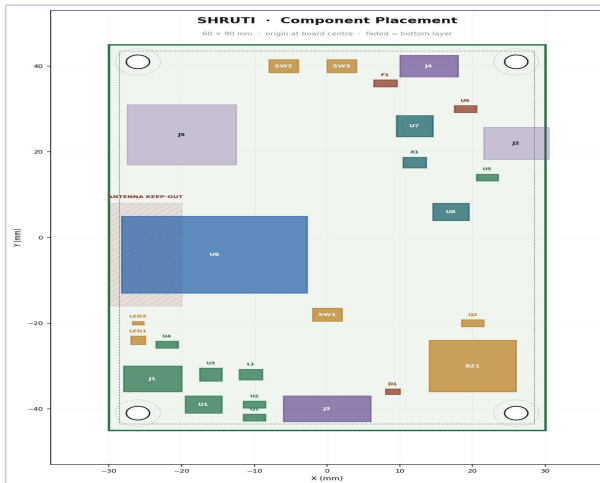
Team STATIC · Smart India Hackathon 2026 · 60 × 90 mm, 4-layer



Signal flow: MCU synthesises f_1 and f_2 → codec DAC → SEPARATE outputs HPL/HPR → two receivers. Emission returns through the microphone → preamp → codec ADC → MCU FFT → PASS / REFER / RETEST.

PCB — SHRUTI main board

60 × 90 mm, 4-layer
ESP32-S3 + TLV320AIC3204
codec, one MCLK domain
Two balanced-armature
receivers on separate 3V3A
rail
Antenna keep-out
respected in placement



FEASIBILITY AND VIABILITY



◆ Analysis of the feasibility of the idea

The noise budget closes — with 13 dB to spare

Mic self-noise 27 dB(A) → -13.0 dB SPL/VHz → -5.3 dB in a 5.86 Hz FFT bin

After 4 s of coherent averaging (23 frames, +13.6 dB): **floor = -18.9 dB SPL**

Target emission = 0 dB SPL, criterion = 6 dB SNR ⇒ **13 dB of margin.**

Electronic noise is therefore NOT the limit — ambient noise and the system's own distortion are, and both are engineered against.

Where the ₹5,000 actually goes — costed, not asserted

2 × balanced-armature receivers	₹1,600	ESP32-S3-WROOM-1 N16R8	₹350
Low-noise microphone	₹700	Preamp, drivers, power, 4-layer PCB	₹1,030
24-bit codec (TLV320AIC3204)	₹450	Probe, enclosure, OLED, tips, cable	₹870

₹5,000 prototype · ₹2,700 at 10,000 units · vs ₹1.85–2.9 lakh imported

How we will prove it works — acceptance criteria set in advance

TIER 1

Bench — no ethics approval needed

System $2f_1-f_2 \leq -15$ dB SPL, and ≤ 1 PASS in 120 repetitions into sealed 0.05 / 0.1 / 0.2 mL cavities

TIER 2

n ≈ 30 normal-hearing adults

PASS/REFER agreement against a clinical ERO-SCAN / Titan, reported as Cohen's kappa plus Bland–Altman on DP amplitude

TIER 3

Neonatal — declared, not claimed

Protocol written and IEC submission prepared with a clinical partner; we will not claim newborn data we could not lawfully have obtained

IMPACT AND BENEFITS



◆ Potential impact on the target audience

1.15–1.38

lakh infants born hard of hearing in India every year

24 mo → <1 mo

age at detection, moving intervention inside the < 6-month window

74x

lower capital cost to equip a facility with a screener

₹493 crore

saved to equip 25,000 public facilities nationwide

Who it reaches, and what changes for them

- Newborns & families — deafness caught in weeks, not years; spoken language remains achievable
- ASHAs, ANMs & staff nurses — an objective test they can run themselves, bypassing the 1:5,00,000 audiologist ratio
- PHCs, CHCs & district hospitals — one screener per delivery point instead of one per district
- NPPCD / RBSK programme managers — live coverage, refer-rate and follow-up data where none exists today
- Audiologists & ENT departments — scarce ABR capacity is spent on pre-filtered referrals, not walk-in guesswork

Social · Economic · Environmental

SOCIAL Early intervention preserves speech, schooling and lifetime earning capacity; screening reaches outreach camps and home births.

ECONOMIC ₹500 crore → ₹6.75 crore to equip 25,000 facilities. Funded from NPPCD/RBSK equipment lines that already exist — an existing line item, not a new ask. Avoids the lifetime cost of special education and lost productivity.

ENVIRONMENTAL Repairable, battery-powered, domestically manufacturable; the only consumable is a ₹5 ear tip. Import substitution for a device currently bought abroad.

THE MULTIPLIER — closing the loop that every screening programme in the world leaks through



RESEARCH AND REFERENCES



◆ Details / Links of the reference and research work

- 1 **Infant Hearing Screening in India: Current Status and Way Forward**
5–6 per 1,000 neonates; detection after age 2; machine purchase cost as the stated NPPCD barrier
[pmc.ncbi.nlm.nih.gov/articles/PMC4689099/](https://pubmed.ncbi.nlm.nih.gov/articles/PMC4689099/)
- 2 **Perspectives of newborn hearing screening in resource-constrained settings**
38% of medical colleges screen; audiologist ratio ~1:5,00,000; NPPCD 2006 and RBSK 2013
[pmc.ncbi.nlm.nih.gov/articles/PMC7691834/](https://pubmed.ncbi.nlm.nih.gov/articles/PMC7691834/)
- 3 **Making a Difference from Day One: Universal Neonatal Hearing Screening**
Intervention most effective before six months of age
[pmc.ncbi.nlm.nih.gov/articles/PMC11674813/](https://pubmed.ncbi.nlm.nih.gov/articles/PMC11674813/)
- 4 **Egyptian National Newborn Hearing Screening Programme — 4-year outcomes**
The five programme quality metrics, including follow-up attendance
[pmc.ncbi.nlm.nih.gov/articles/PMC12641659/](https://pubmed.ncbi.nlm.nih.gov/articles/PMC12641659/)
- 5 **Gorga et al. — Identification of neonatal hearing impairment: DPOAEs (Ear & Hearing, 2000)**
SNR best at 3–4 kHz; poorest performance and longest test time at 1 kHz
pubmed.ncbi.nlm.nih.gov/11059701/
- 6 **Stakeholders' perspective for improved UNHS uptake in Odisha (J. Tropical Pediatrics, 2021)**
Prevalence and RBSK scope for congenital hearing loss in India
academic.oup.com/tropej/article/67/3/fmaa062/5905598